
Analyzing Skills and Salary Trends in the Artificial Intelligence Job Market

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01:198:439:05 – Intro to Data Science

Abstract

Artificial Intelligence is one of the fastest growing areas in technology, and many students are interested in understanding what kinds of jobs exist in this field and which skills employers value the most. In this project, I will analyze an AI job market dataset from Kaggle to study skill demand and salary trends in artificial intelligence related jobs. The dataset includes information such as job title, salary, experience level, company size, required skills, education, and remote work information. Using data science methods such as data cleaning, exploratory data analysis, visualization, regression, and clustering, this project aims to identify the most common skills in AI jobs and examine how job features influence salary. The results may help students better understand current AI career trends and prepare for future job opportunities.

1 Research Question and Motivation

Research Question

The main research question for this project is: What skills are most demanded in Artificial Intelligence jobs, and how do job characteristics such as experience level, company size, and remote work influence salary?

Motivation

Artificial Intelligence is one of the fastest growing areas in technology and many students in computer science and data science want to work in this field. However, it is sometimes difficult for students to understand what kinds of jobs exist and what specific skills employers expect.

As a Computer Science and Data Science major, I personally found it challenging to understand which skills and tools are most important for AI-related jobs. Because of this experience, I decided to work on a project that analyzes AI job market data. This project will help explore which skills appear most frequently in job postings and what factors influence salary in AI roles.

Background Knowledge and Prior Work

Many studies and reports have analyzed technology job markets to understand skill demand and salary trends. Job market analysis using data science methods has become common in understanding labor market trends. Platforms such as Kaggle and LinkedIn provide datasets that allow researchers and students to study patterns in job postings.

This project builds on similar work by applying data science techniques such as exploratory data analysis, regression, and clustering to better understand trends in the AI job market.

2 Data Sources

The dataset for this project will come from Kaggle. The dataset link is:

<https://www.kaggle.com/datasets/bismasajjad/global-ai-job-market-and-salary-trends-2025>

This dataset contains information about AI-related jobs including job title, salary, experience level, company size, required skills, industry, education requirements, remote work ratio, and years of experience.

The dataset will be downloaded from Kaggle and analyzed using Python. Data preprocessing will include checking for missing values, removing duplicate records if necessary, cleaning column names, and preparing the skills column for analysis. Since some rows may contain multiple skills in one column, text processing will be used to separate and analyze those skills.

Possible challenges include handling missing values, cleaning text-based skill data, and preparing categorical variables for modeling.

3 Methodology and Analysis Plan

The project will begin with exploratory data analysis to understand patterns in the dataset. This includes examining the distribution of job titles, salary ranges, and experience levels. Visualizations such as bar charts, histograms, and box plots will be created to better understand trends in the data.

Next, the project will analyze the most commonly required skills in AI jobs by counting how frequently different skills appear in the dataset. This will help identify the most demanded technical skills.

Salary analysis will also be performed to study how salary varies across experience level, company size, and remote work options.

Several data science techniques will also be applied. Linear regression may be used to study the relationship between salary and job characteristics. Clustering may be used to group similar jobs based on required skills and other features.

The main tools and libraries used for this project will include Python, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

4 Expected Outcomes and Evaluation

The expected outcome of this project is to identify important patterns in the AI job market. The analysis should show which technical skills appear most often in job postings and how different factors such as experience level and company size influence salary.

The project will also provide visual insights about salary distributions, job roles, and skill demand.

Model performance will be evaluated using common metrics such as R^2 and mean squared error for regression models. Clustering results will be evaluated by examining how well the job groups represent meaningful job patterns.

A possible extension of this project could include comparing AI job trends across different countries or comparing AI jobs with other technology fields.

Overall, this project aims to provide useful insights about AI career trends and help students better understand the skills needed for future jobs in this field.